RAD Chapter 3

Agile Software Development

Simple English Short Notes

Made from the uploaded lecture slides. Focus: exam revision, quick understanding, and short answers.

Main topics: RAD vs Agile, Agile vs Waterfall, Agile Manifesto, Scrum, XP, Lean, Kanban.

1. Chapter Overview

This chapter explains Agile Software Development. Agile is a flexible way to build software step by step. The team works in small cycles, gets feedback often, and improves the product continuously.

- RAD and Agile both use iterative development.
- Agile is usually organized into short cycles called sprints.
- Scrum, Extreme Programming, Lean, and Kanban are common Agile frameworks.
- Agile focuses on working software, teamwork, customer feedback, and change handling.

2. RAD vs Agile

RAD and Agile are similar because both avoid long, rigid development. Both try to deliver software faster using repeated feedback.

RAD	Agile
Focuses on quickly building a working model or prototype.	Focuses on breaking work into small sprints.
May show a working model to the client in the middle of development.	Usually shows a completed feature/product increment after a sprint.
Often builds easy parts first to show progress quickly.	Often handles difficult or important features early.
Functionality is more important than UI/UX at first.	UI/UX is considered while building the product.

Easy memory: RAD = fast prototype. Agile = small sprint-based development.

3. Agile vs Waterfall

Waterfall is a step-by-step method. One stage is completed before the next stage starts. Agile is flexible and incremental. It allows feedback and change during development.

Agile	Waterfall
Incremental and iterative approach.	Sequential design approach.
Progress is measured by delivered working features.	Progress is measured by completed documents/artifacts.
Requirements can change after development starts.	Requirements are difficult to change after development starts.
High customer interaction.	Less customer interaction.
Testing is reviewed after each sprint.	Testing is reviewed after full development.
Suitable when requirements may change.	Suitable when requirements are clear and stable.

4. Agile Manifesto

The Agile Manifesto was published in 2001 by 17 software developers. It gives the main values and principles for Agile software development.

4.1 Four Values of Agile Manifesto

1. **Individuals and interactions** over processes and tools.
2. **Working software** over comprehensive documentation.
3. **Customer collaboration** over contract negotiation.

4. **Responding to change** over following a fixed plan.

Important: Agile does not say tools, documents, contracts, and plans are useless. It says people, working software, collaboration, and change response are more important.

4.2 The 12 Agile Principles - Simple Version

1. Satisfy the customer by early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development.
3. Deliver working software frequently.
4. Business people and developers should work together daily.
5. Build projects around motivated people.
6. Face-to-face conversation is the best communication method.
7. Working software is the main measure of progress.
8. Agile supports sustainable development pace.
9. Technical excellence and good design improve agility.
10. Simplicity means avoiding unnecessary work.
11. Best designs come from self-organizing teams.
12. Teams regularly reflect and improve their work.

5. Agile Frameworks

An Agile framework is a structured way to apply Agile ideas in real projects.

- **Scrum** - work is done in sprints with defined roles and meetings.
- **Extreme Programming (XP)** - focuses strongly on coding quality and frequent testing.
- **Lean Software Development** - focuses on customer value and removing waste.
- **Kanban** - uses a visual board to manage workflow and limit work in progress.
- Other frameworks include Feature Driven Development, Adaptive System Development, Dynamic Systems Development Method, and Crystal Clear.

6. Scrum

Scrum is one of the most popular Agile frameworks. Work is done in a fixed time period called a sprint. Usually, a sprint lasts 1 to 4 weeks.

6.1 Scrum Process

1. Product owner creates and orders the product backlog.
2. In sprint planning, the team selects top backlog items and creates the sprint backlog.
3. The team works during the sprint and discusses progress in daily scrum meetings.
4. At the end of the sprint, the team shows completed work in the sprint review.
5. The team improves its process in the sprint retrospective.
6. Product backlog refinement keeps future backlog items clear and ready.

6.2 Scrum Roles

Role	Simple Meaning	Main Duties
Product Owner	Client/business representative.	Defines product features, prioritizes backlog, decides release content, accepts or rejects work.
Scrum Master	Coach and facilitator.	Protects team, supports Scrum values, removes blockers, improves team effectiveness.
Team / Developers	People who build the product.	Self-organizing, cross-functional team of about 5-9 members. Developers and testers work together.

6.3 Scrum Activities

Activity	When	Purpose
Sprint Planning	Beginning of sprint	Plan what to build and how to build it. Output: sprint goal, sprint backlog, tasks, estimates.
Daily Scrum	Every day	15-minute meeting. Team discusses yesterday work, today plan, and blockers.
Sprint Review	Last day of sprint	Show completed working product and get feedback.
Sprint Retrospective	End of sprint	Discuss how to improve people, process, and culture.
Product Backlog Refinement	Between sprints / continuous	Clarify, reorder, add, remove, or modify backlog items for future sprints.

6.4 Scrum Artifacts

- **Product Backlog:** Ordered list of product ideas, requirements, and features. Product owner maintains it.
- **Sprint Backlog:** Product backlog items selected for the current sprint with the team plan.
- **Product Increment:** Working product output from a sprint. It must meet quality standards and be acceptable to the product owner.

6.5 Scrum Information Radiators

- **Task Board:** Shows work status using columns like backlog, to-do, in progress, and done.
- **Burndown/Burnup Chart:** Shows remaining work or progress against time. It helps the team track sprint progress.

7. Extreme Programming (XP)

Extreme Programming is an Agile method that focuses strongly on software engineering practices. It takes good development practices to an extreme level. Example: code review becomes continuous through pair programming.

7.1 Five XP Values

1. Communication
2. Simplicity
3. Feedback
4. Courage
5. Respect

7.2 Twelve XP Practices

Group	Practices
Fast feedback	Test driven development, on-site customer, pair programming.
Continuous process	Continuous integration, code refactoring, short releases.
Shared understanding	User stories, simple design, system metaphor, collective ownership, coding standard.
Developer welfare	40-hour workweek.

7.3 Important XP Practice Meanings

- **User stories:** Requirements are written as simple scenarios or story cards.
- **Short releases:** Software is released frequently in short time gaps.
- **System metaphor:** Common naming and shared understanding of the system.
- **Collective ownership:** Whole team owns the code, not one person only.
- **Coding standard:** Same coding style helps collaboration.
- **Simple design:** Build the simplest solution that works.
- **Refactoring:** Improve code structure without changing behavior.
- **Pair programming:** Two programmers work together.
- **On-site customer:** Customer is available to guide the team.
- **Test-first development:** Write tests before coding.
- **Continuous integration:** Integrate code many times, not only at the end.

7.4 Advantages and Disadvantages of XP

Advantages	Disadvantages
Good for small and medium projects. Iterative development. Frequent delivery and testing. High-quality code. Strong user involvement.	Hard to scale for large projects. Needs skilled test writing. Requires full-time customer involvement. Documentation may be less.

8. Lean Software Development (LSD)

Lean Software Development is an Agile framework based on Lean manufacturing ideas from Toyota. It focuses on delivering maximum customer value with minimum waste. It is also linked with MVP - Minimum Viable Product.

8.1 Key Principles of Lean Software Development

1. **Eliminate waste:** Remove unnecessary work, extra features, waiting, defects, task switching, and handoffs.
2. **Build quality:** Focus on quality and customer value from the beginning.
3. **Amplify learning:** Development is continuous learning.
4. **Defer commitment:** Make decisions after enough analysis; keep software flexible.
5. **Fast delivery:** Deliver small batches quickly and get early feedback.
6. **Respect teamwork:** Empower the team and value people.
7. **Optimize the whole system:** Improve the full workflow, not only one part.

8.2 Seven Wastes in Lean

1 Partially done work
2 Extra features
3 Revisiting decisions
4 Task switching
5 Waiting
6 Handoffs
7 Defects

8.3 Advantages and Weaknesses of Lean

Advantages	Weaknesses
Saves time by removing unnecessary stages. Focuses on MVP and essential functions. Improves team involvement and workflow.	Depends heavily on the team. Difficult to keep speed. Decision-making becomes hard if customer requirements are unclear.

9. Kanban

Kanban is a flexible Agile method that uses a visual board to manage work. The word Kanban means visual board or sign board in Japanese.

9.1 Kanban Board

- Columns show stages of work, such as To Do, In Progress, and Done.
- Cards represent individual tasks.
- WIP limits restrict the maximum number of tasks in a stage.
- Swimlanes visually separate different types of work.

9.2 Kanban Practices

1 Visualize the workflow.
2 Limit work in progress.
3 Focus on smooth flow.
4 Continuously improve.
5 Take feedback using reviews, reports, risk meetings, and standups.
6 Improve collaboratively and experiment.

9.3 Advantages and Disadvantages of Kanban

Advantages	Disadvantages
Helps identify issues. Flexible. Saves time. Empowers teams. Balances productivity.	Less effective with shared resources. Demand changes may create problems. It does not remove all variability. Quality issues can still happen.

10. Quick Comparison

Method	Main Idea	Best For
RAD	Build a quick prototype/model.	Projects needing fast visible progress and user feedback.

Method	Main Idea	Best For
Agile	Build software in small cycles with feedback.	Changing requirements and continuous delivery.
Waterfall	Complete stages one by one.	Clear and fixed requirements.
Scrum	Sprint-based Agile framework with roles and ceremonies.	Team projects needing structure and regular delivery.
XP	Agile with strong coding practices.	Small/medium projects needing high code quality.
Lean	Deliver value by removing waste.	MVP and efficiency-focused development.
Kanban	Visual workflow with WIP limits.	Continuous flow of tasks and flexible teams.

11. Exam-Ready Short Answers

Q1. What are the four values of Agile Manifesto?

1. Individuals and interactions over processes and tools.
2. Working software over comprehensive documentation.
3. Customer collaboration over contract negotiation.
4. Responding to change over following a plan.

Q2. List five Scrum activities.

1. Sprint planning
2. Daily scrum
3. Sprint review
4. Sprint retrospective
5. Product backlog refinement

Q3. List three Scrum artifacts.

1. Product backlog
2. Sprint backlog
3. Product increment

Q4. What are the values of XP?

1. Communication
2. Simplicity
3. Feedback
4. Courage
5. Respect

Q5. What is a sprint?

A sprint is a fixed time period in Scrum, usually 1 to 4 weeks, during which the team builds a selected set of features.

Q6. What is a product backlog?

Product backlog is an ordered list of product features, ideas, and requirements. The product owner maintains and prioritizes it.

Q7. What is a sprint backlog?

Sprint backlog is the selected product backlog items and tasks that the team plans to complete in the current sprint.

Q8. What is a product increment?

A product increment is the working product output produced at the end of a sprint. It must meet the definition of done and be acceptable to the product owner.

Q9. What is WIP limit in Kanban?

WIP limit means Work-In-Progress limit. It controls the maximum number of tasks allowed in a workflow stage to avoid overload.

Q10. What is MVP in Lean?

MVP means Minimum Viable Product. It is a small version of the product with only essential features, released quickly to get feedback.

12. One-Page Memory Sheet

Agile = small steps + feedback + change friendly + working software.

- RAD: quick prototype.
- Agile: sprint/increment based flexible development.
- Waterfall: sequential, less flexible.
- Scrum roles: Product Owner, Scrum Master, Team.
- Scrum activities: Planning, Daily Scrum, Review, Retrospective, Backlog Refinement.
- Scrum artifacts: Product Backlog, Sprint Backlog, Product Increment.
- XP values: Communication, Simplicity, Feedback, Courage, Respect.
- Lean principles: remove waste, build quality, learn, defer decisions, deliver fast, respect team, optimize whole.
- Kanban: visual board + cards + WIP limits + continuous improvement.

End of Chapter 3 Short Notes